

AL58 - RESEARCH METHODOLOGY AND INTELLECTUAL PROPERTY RIGHTS

UNIT - I RESEARCH PROPOSAL & LITERATURE SCOPING WORKSHEET

SECTION 1: RESEARCH CONCEPTUALIZATION & OBJECTIVES

Primary Research Objective: (Select one main goal for your research work)

- Gain new insights into a phenomenon
- Accurately portray characteristics of a system or group
- Determine occurrence frequency or associations
- Test a causal relationship between variables

Research Classification: (Select one across each of these pairs)

- Approach: Descriptive OR Analytical
- Application: Applied (action-oriented) OR Fundamental (pure theory)
- Data Type: Quantitative (measurable numbers) OR Qualitative (qualities/phenomena)
- Method: Conceptual (abstract theory) OR Empirical (observation/experiment)

SECTION 2: ETHICAL COMPLIANCE & INTEGRITY SCOPING

Ethical Principles to Maintain: (How will you ensure objectivity and care in execution?)

Misconduct Prevention Plan: (Define strategies to avoid these specific pitfalls)

- Fabrication Risk: (How will you ensure raw data is recorded accurately ?)
- Falsification Risk: (How will you ensure experimental results are not manipulated ?)
- Plagiarism Risk: (What similarity check tools will ensure ideas and text are credited?)

SECTION 3: LITERATURE REVIEW & PRIOR ART SYNTHESIS

Bibliographic Databases Used: (e.g., IEEE Xplore, Web of Science, Google Scholar)

Synthesis of Prior Art:

- Existing Knowledge State: (What is already known from current publications?)
- Identified Shortcoming / Inconsistency: (What loopholes, unaddressed challenges, or contradictory results exist in prior work?)
- New Knowledge Contribution: (What specific missing part will your research create?)
- Critical Assessment: (What weakness or limitation was found in prior models?)
- Creative Extension: (What novel modification can be adapted for your study?)

SECTION 4: CITATIONS, ATTRIBUTION & KNOWLEDGE FLOW

Impact of Search Attributes:

- Title Strategy: (Draft an informative, concise title to maximize discoverability)
- Key Search Keywords: (List 4-5 core domain keywords used by indexing engines)
- Citation Style: (e.g., IEEE style for Computer Science)
- Funding Agencies / Grants to Acknowledge: (Name computing grants, lab facilities or funding schemes)
- Technical / Peer Contributions: (List mentors, dataset creators or lab members assisting the work)